

(7.)

$$U = K \left[\frac{q_1 q_2}{d_{12}} + \frac{q_1 q_3}{d_{13}} + \frac{q_2 q_3}{d_{23}} \right]$$

$$U = 9 \cdot 10^9 \left[\frac{2 \cdot 10^{-6} \cdot -6 \cdot 10^{-6}}{3} + \frac{2 \cdot 10^{-6} \cdot 3 \cdot 10^{-6}}{4} + \frac{3 \cdot 10^{-6} \cdot (-6 \cdot 10^{-6})}{5} \right]$$

$$U = -0,0549 \text{ [J]}$$

$$\begin{aligned} d_1 &= (2,1) - (0,0) \\ &= \sqrt{2^2 + 1^2} \\ &= \sqrt{5} \end{aligned}$$

$$\begin{aligned} d_1 &= (4,2) - (0,0) \\ &= \sqrt{4^2 + 2^2} \\ &= \sqrt{20} \end{aligned}$$

$$\begin{aligned} d_2 &= (2,1) - (4,0) \\ &= \sqrt{2^2 + 1^2} \\ &= \sqrt{5} \end{aligned}$$

$$\begin{aligned} d_2 &= (4,2) - (4,0) \\ &= \sqrt{2^2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} d_3 &= (2,1) - (0,3) \\ &= \sqrt{2^2 + 2^2} \\ &= \sqrt{8} \end{aligned}$$

$$\begin{aligned} d_3 &= (4,2) - (0,3) \\ &= \sqrt{4^2 + 1^2} \\ &= \sqrt{17} \end{aligned}$$

$$\begin{aligned} V_{z1} &= k \left[\frac{2\mu C}{\sqrt{5}} + \frac{3\mu C}{\sqrt{5}} - \frac{6\mu C}{\sqrt{8}} \right] \\ &= 1032,72 \text{ [V]} \end{aligned}$$

$$\begin{aligned} V_{z2} &= k \left[\frac{2\mu C}{\sqrt{20}} + \frac{3\mu C}{2} - \frac{6\mu C}{\sqrt{17}} \right] \\ &= 4427,99 \text{ [V]} \end{aligned}$$